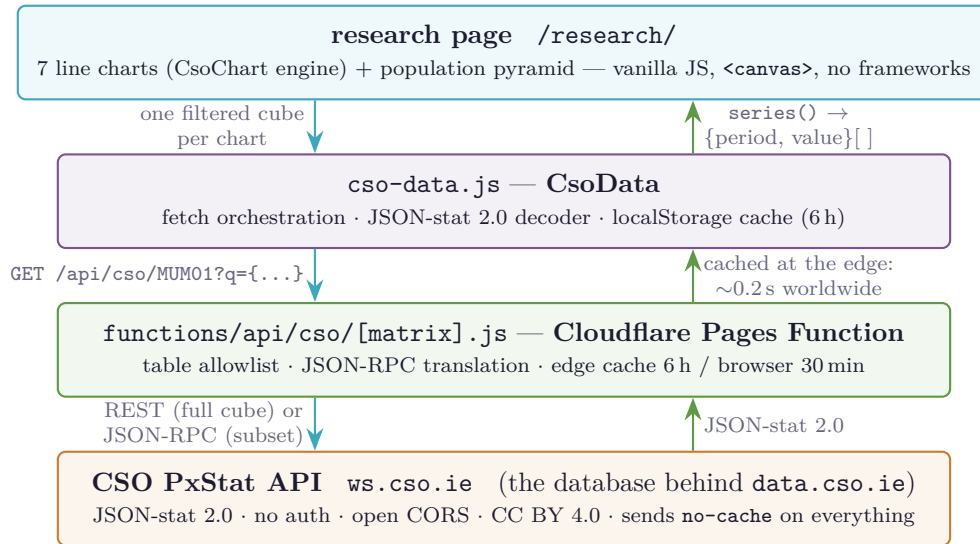


Icecon.ie × CSO PxStat

The live data pipeline, at a glance

July 2026 — every fact verified against the live API

1 The whole system



Deploy is *push to main*: Cloudflare Pages rebuilds the static site and the proxy function together, ~1 minute, zero configuration.

2 Why a proxy? The four-level fallback

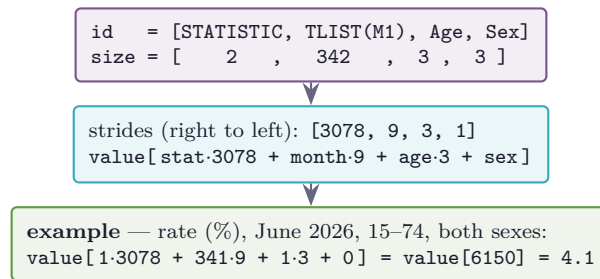
The CSO sends `cache-control: no-cache` on every response, so without our own caching every student pays a cold round trip to CSO origin. The page also never breaks: each level below catches the failure of the one above.



Level 2 also covers local development (`python3 -m http.server` runs no Pages Functions), so the same code works everywhere unchanged.

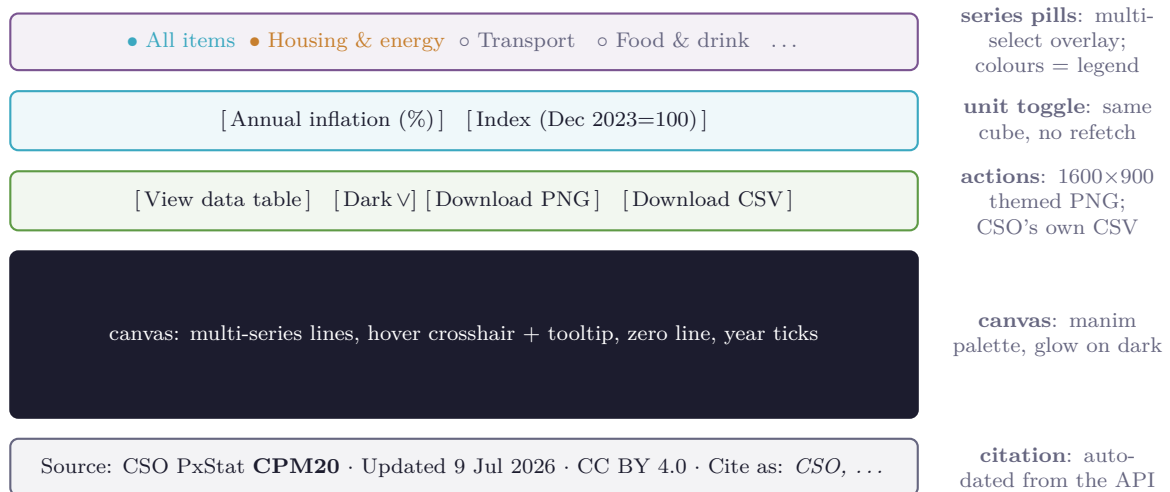
3 Decoding JSON-stat: the one piece of real arithmetic

A cube arrives as one flat `value[]` array plus its dimensions. The flat index is row-major over the `id` order — exactly like nested loops:



`CsoData.series(ds, fixed)` does this for a whole time axis: pin every non-time dimension, walk the time categories, return `[{code, label, value}]`. One PxStat quirk: `category.index` is an *array* of codes (Eurostat uses an object) — the decoder handles both.

4 Anatomy of a chart block



The engine builds all of this from a ~20-line config; the pill list comes from live metadata, so new CSO categories appear automatically.

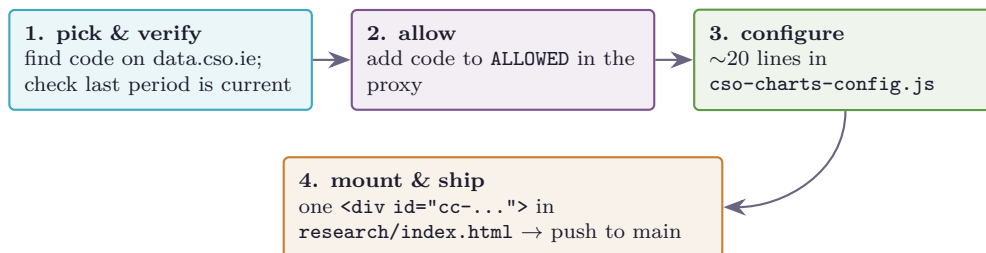
5 The seventeen visualisations and their tables

Three renderers share one data layer: **lines** (`CsoChart`), **donuts** with a year slider (`CsoPie`), and the **regional map** (`CsoMap`).

Chart	Table	Span	Pills / toggle / slider
Unemployment	MUM01	1998– (monthly)	3 ages / % vs '000s
Inflation (CPI)	CPM20	1996– (monthly)	14 commodity groups / % vs index
GDP–GNP–GNI–GNI*	NA001	1995– (annual)	4 aggregates, €bn
House prices	HPM09	2005– (monthly)	6 regions/types / index vs %
Dwelling completions	NDQ01	2011– (quarterly)	4 house types / raw vs SA
Government debt	GFQ12	2000– (quarterly)	single series, % of GDP
Merchandise trade	TSM01	1970– (monthly)	exports, imports, surplus (±SA)
Population pyramid	PEA11	1926– (year slider)	single year of age × sex
Tax revenue (donut)	ITXS01	1995– (year slider)	top-9 tax heads + Other
Gov. rev/exp/deficit	GFA01	1995– (annual)	revenue, expenditure, B9
Social protection (donut)	SPEA02	2000– (year slider)	9 benefit functions
Balance of payments	BPQ15	2002– (quarterly)	current acct, goods, services, income
Trade in services	BPA04	2003– (annual)	8 components / exports vs imports
Industrial production	MIM05	2015– (monthly)	modern vs traditional sector
Retail sales	RSM08	2021– (monthly)	6 business types / volume vs value
Services index	MSI03	2021– (monthly)	6 sectors / volume vs value
Housing map	NDQ08	2011– (map + slider)	8 NUTS3 regions, GISCO geometry

Trap to remember: the PxStat catalogue keeps discontinued tables with no flag (CPM13 ends 2016, HPM01 ends 2019, ...). Before adopting any table, check the last period in its time dimension via `ReadMetadata`.

6 Adding chart #9



No drawing code, no fetch code, no DOM code — the engine owns all of it. Full prose reference: `docs/cso-data-pipeline.md`.